

FUSION HIGH EDUCATIONAL ECOSYSTEM

Comprehensive Technical Architecture & Functional Documentation Guide

Next-Generation Multi-Tenant School Management & DBE CAPS Cloud Infrastructure
Covering Limpopo (Capricorn South / Mankweng) & Gauteng (Tshwane South & West Districts)

DOCUMENT SPECIFICATIONS & METADATA

- System Release Version: v2.1.0 Enterprise Production
- Target Database Engine: PostgreSQL 15+ (Relational / Schema-Isolated)
- Curriculum Standard: South African Department of Basic Education (CAPS & SASAMS)
- Executive Administrator: Dr. Tshepo Makola (SuperAdmin Executive)
- Supported Client Roles: Main SuperAdmin, School SubAdmin, Educator, Parent, Learner
- Official Publication Date: August 2026

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 - Digital School Fees, Online Payments & Invoicing
 - Grade 12 Matric Pass Rate & Endorsement Projector
- Security, Role-Based Access Control (RBAC) & DBE Compliance

1. Multi-Tenant Architecture & Governance

The platform operates on a robust Multi-Tenant Model designed to serve 12 independent high schools while maintaining strict data isolation. The hierarchy distinguishes between the Main Executive Administrator (SuperAdmin) and dedicated School SubAdmins.

- Main Executive Administrator (is_superadmin = true): Holds system-wide visibility across all institutions. Can appoint SubAdmins, schedule inter-school tournaments, view global revenue and attendance averages, and switch between school views.
- Dedicated School SubAdmin (is_superadmin = false, role = "admin", school_id = X): Bound strictly to their designated school campus. Manages teachers, assigns subject workloads, registers parents, enrolls learners, generates conflict-free timetables, and compiles term report cards for their school only.

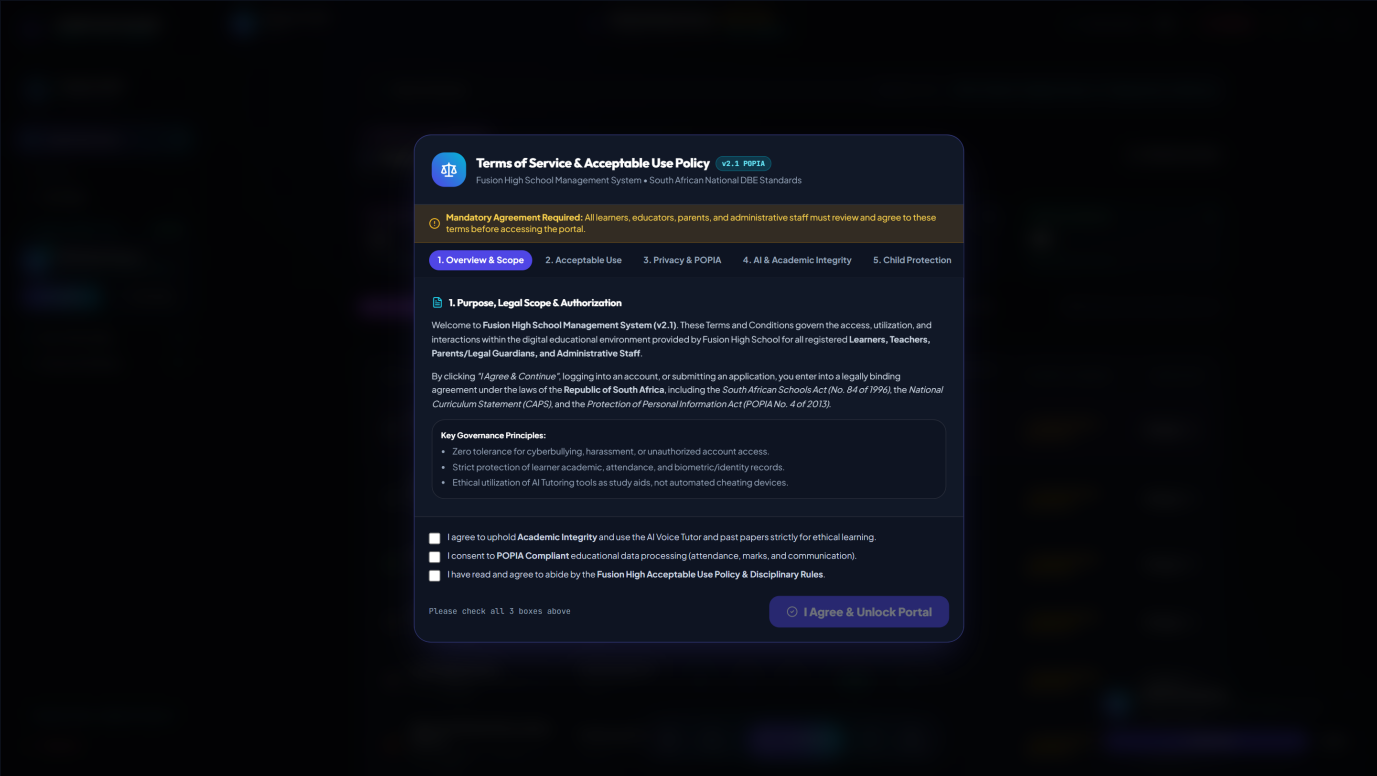


Figure 1: Multi-School Command Center displaying comparative side-by-side metrics across all 12 institutions.

2. Inter-School Collaboration & League

To foster district spirit and healthy academic/athletic rivalry, the system includes an Inter-School Competitions & Derby League. Schools across Mankweng and Pretoria participate in organized matches and academic tournaments with live scoreboards.

- Supported Activities: Soccer, Netball, Rugby, Chess, Athletics, Mathematics Olympiads, Science Fairs, and Debates.
- Live Scoreboard: Displays home/away crests, official scores, match highlights, and trophy titles.
- Standings Leaderboard: Automated points table calculating Wins (3 pts), Draws (1 pt), Losses (0 pts), and Trophies won.

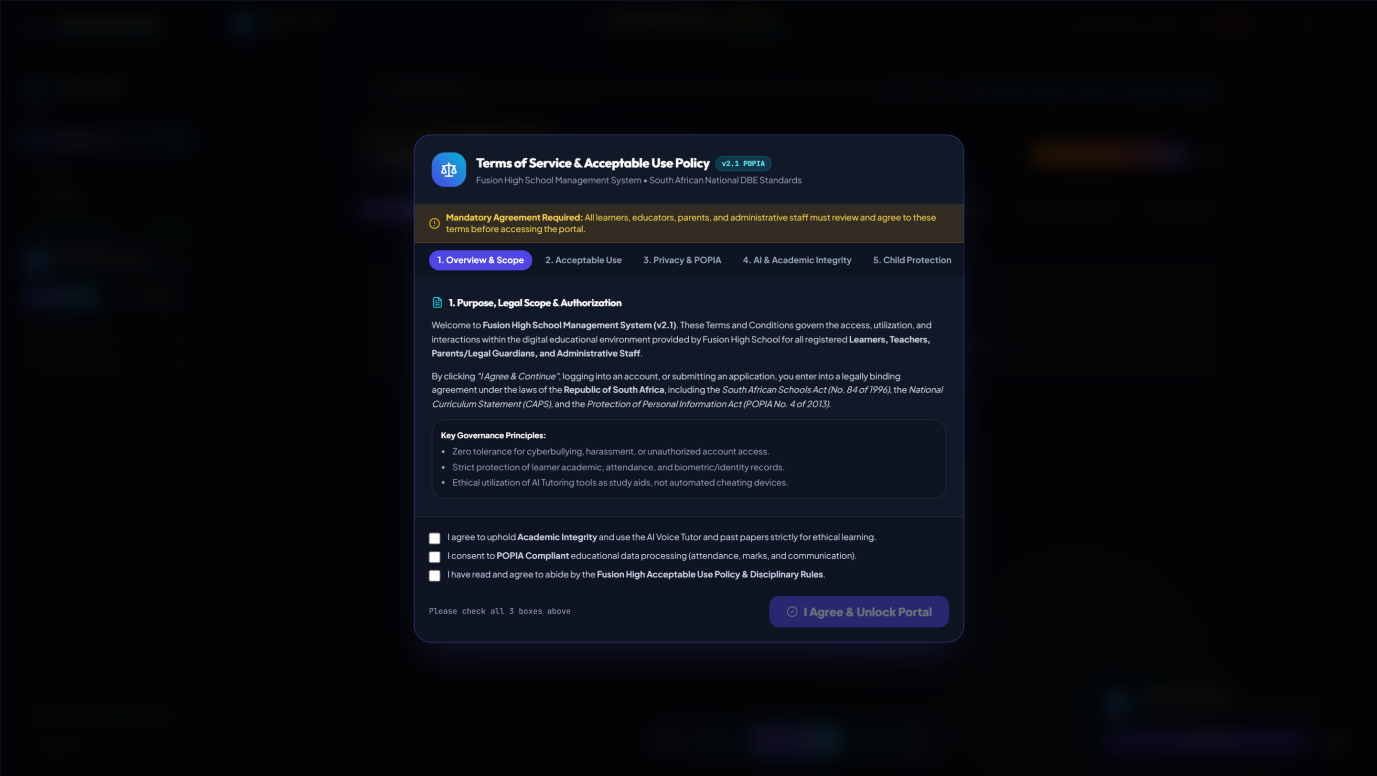


Figure 2: Inter-School Match Day Hub & Championship Leaderboard.

3. Parent-Educator Consultations & CAPS Reports

Effective family-educator collaboration and transparent academic reporting are central to student achievement.

- 20-Minute Consultation Scheduler: Parents select their child and educator to book open time slots. Automated email notices are dispatched to both parties with meeting agendas.
- Official DBE CAPS Report Cards: Aggregates real SBA continuous assessment marks, applies the 7-Point Achievement Scale (Level 1–7), adds teacher remarks, and batch emails signed academic report cards to parents.

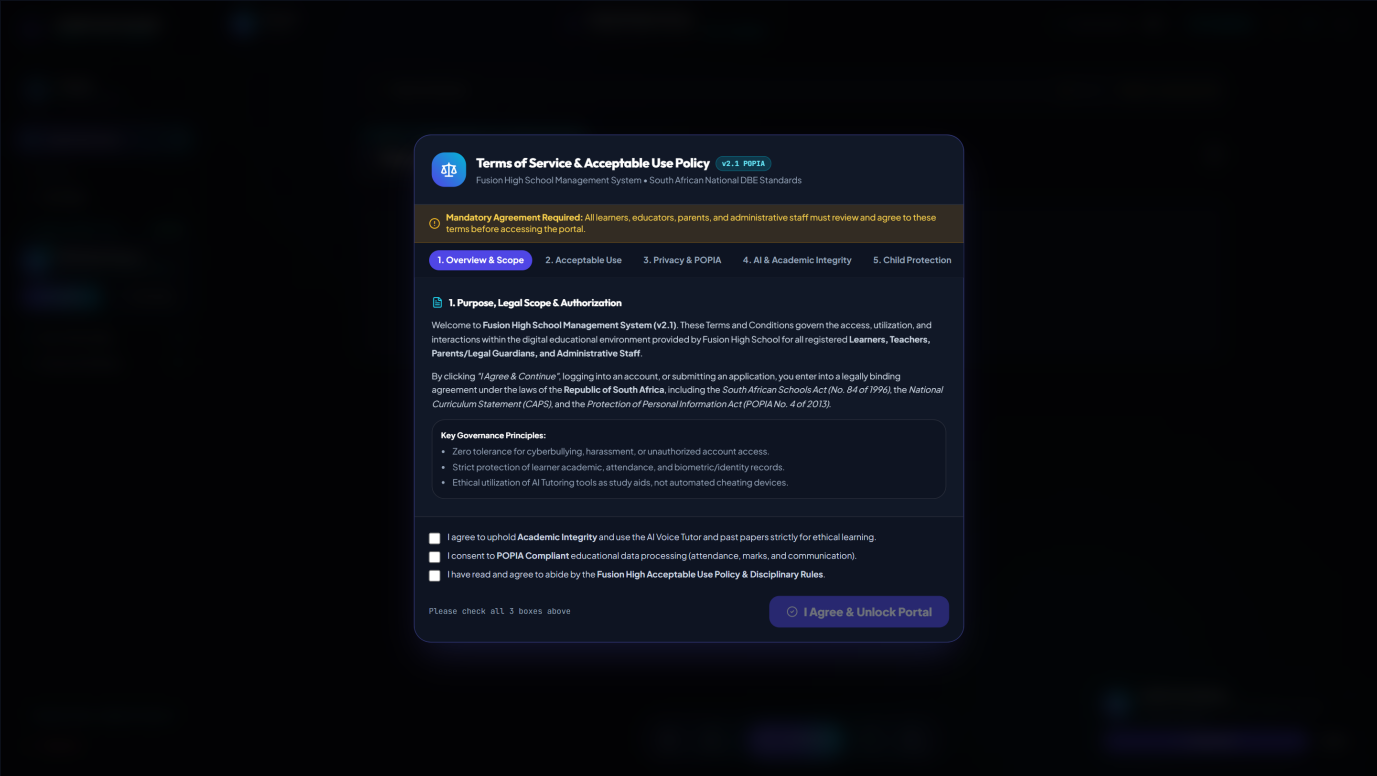


Figure 3: Parent-Educator 20-Minute Academic Consultation Scheduler with Automated Email Confirmations.

4. School-Aware AI Academic Specialist

The AI Academic Tutor integrates strict multi-tenant grounding based on the authenticated learner's registered school campus.

- Institutional Sensing: The assistant automatically detects the learner's school (e.g. Flavius Mareka Secondary School, Mountainview, Fusion High) and incorporates school circuit, motto, and provincial standards into the system prompt.
- Zero Cross-School Leakage: The AI represents the user's school exclusively without mentioning or confusing other institutions.
- Past-Paper Examination Mode: Quizzes students with CAPS-aligned questions, evaluates submitted answers, and generates step-by-step marking memorandums.
- Official Language Specialization: Supports fluent explanations across all 11 official South African languages.

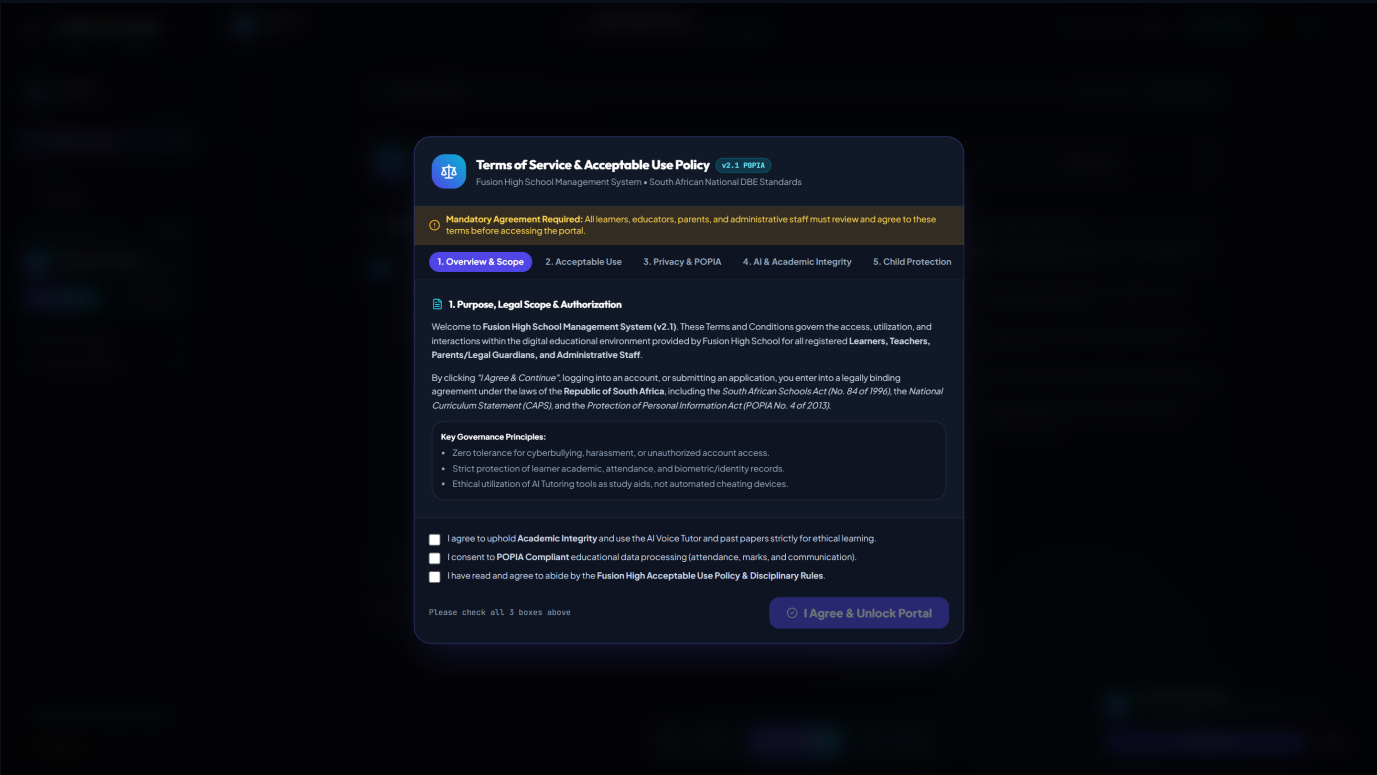


Figure 4: School-Aware AI Subject Specialist providing CAPS syllabus revision.